
PHYSICS

9702/42

Paper 4 A Level Structured Questions

October/November 2018

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **14** printed pages.



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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	force per unit mass	B1
1(a)(ii)	acceleration = F/m , field strength = F/m , so equal	B1
1(b)	smooth curve between R and $4R$ with negative gradient of decreasing magnitude	B1
	line passing through $(R, 1.00g)$ and $(2R, 0.25g)$	B1
	line ending at $(4R, 0.0625g)$	B1
1(c)	$M = (4/3 \times \pi R^3)\rho$	C1
	$g = GM/(2R)^2$	C1
	$g = \frac{1}{3} \times 6.67 \times 10^{-11} \times \pi \times 3.4 \times 10^6 \times 4.0 \times 10^3$ $= 0.95 \text{ ms}^{-2}$	A1

Question	Answer	Marks
2(a)	gas that obeys equation $pV = \text{constant} \times T$	M1
	symbols p , V and T explained	A1
2(b)(i)	$pV = \frac{1}{3} Nm\langle c^2 \rangle$ and $M = Nm$ (and so) $p = \frac{1}{3} \rho \langle c^2 \rangle$	C1
	$2.12 \times 10^7 = \frac{1}{3} \times [3.20 / (1.84 \times 10^{-2})] \times \langle c^2 \rangle$	C1
	$c_{\text{r.m.s.}} = 605 \text{ m s}^{-1}$	A1
2(b)(ii)	1. $pV = nRT$ and $T = (22 + 273) \text{ K}$	C1
	$n = (2.12 \times 10^7 \times 1.84 \times 10^{-2}) / (8.31 \times 295)$ $= 159 \text{ mol}$	A1
	2. mass = $3.20 / (159 \times 6.02 \times 10^{23})$ or mass = $[2 \times (3/2) \times 1.38 \times 10^{-23} \times 295] / 605^2$	C1
	mass = $3.34 \times 10^{-26} \text{ kg}$	A1
2(c)	$A = (3.34 \times 10^{-26}) / (1.66 \times 10^{-27})$ $= 20$	A1

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Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause change of state)	B1
	(energy transfer during) change of state between solid and liquid at constant temperature	B1
3(b)(i)	Any one from: • rate of increase in mass (of beaker and water) is constant • level of water rises at a constant rate • volume of water (in beaker) increases at a constant rate • constant time between drops • constant rate of dripping	B1
3(b)(ii)	(electrical power supplied =) 12.8×4.60 (= 58.9 W)	C1
	(rate of transfer to ice =) $[(185.0 - 121.5) \times 332] / [5.00 \times 60]$ (= 70.3 W)	C1
	1. rate = 70.3 W	A1
	2. rate = 70.3 – 58.9 = 11.4 W	A1

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Question	Answer	Marks
4(a)	(defining equation of s.h.m. is) $a = -kx$ where k is a constant or $a \propto -x$	B1
	g and L are constant (so $a \propto -x$ and hence s.h.m.)	B1
4(b)	$T = 0.50 \text{ s}$ and $T = 2\pi / \omega$	C1
	$\omega^2 = 2g/L$	C1
	$L = (2 \times 9.81 \times 0.50^2) / 4\pi^2$ $= 0.12 \text{ m}$	A1
4(c)(i)	Any one from: - viscosity of liquid - friction within the liquid - viscous drag - friction/resistance between walls of tube and liquid	B1
4(c)(ii)	(maximum) $\text{KE} = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy = $\frac{1}{2}m\omega^2 x_0^2$	C1
	ratio = $(1.3/2.0)^2$ $= 0.42$	A1

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Question	Answer	Marks
5(a)	amplitude of carrier (wave) varies	B1
	variation in synchrony with displacement of information signal	B1
5(b)(i)	wavelength = $(3.0 \times 10^8)/(900 \times 10^3)$ = 3.3×10^2 m	A1
5(b)(ii)	amplitude varies (continuously) between a maximum and a minimum	B1
	variations repeat 5000 times each second or variations repeat every 0.2 ms or variations above and below 4.0 V	B1
5(b)(iii)	10000 Hz	A1
5(c)(i)	Any two from: • (orbit is) above the Equator • (orbit is) from west to east/same direction as Earth's rotation • orbit is circular/orbit has a particular radius	B2
5(c)(ii)	1. minimal reflection/absorption/attenuation by <u>atmosphere</u> or maximum penetration of/transmission through <u>atmosphere</u>	B1
	2. uplink signal is greatly attenuated/must be greatly amplified	B1
	prevents downlink signal swamping the uplink signal	B1

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Question	Answer	Marks
6(a)(i)	work done per unit charge	B1
	work done moving positive charge from infinity (to the point)	B1
6(a)(ii)	field strength = potential gradient	M1
	negative sign included or directions discussed	A1
6(b)	horizontal straight lines, at non-zero potential, within the spheres	B1
	magnitude of potential greater at surface of sphere A than at surface of sphere B	B1
	concave curve between A and B, with a minimum nearer to B	B1
	lines show V <u>positive</u> all the way from 0 to D	B1

Question	Answer	Marks
7(a)	$R/R_T = 2.4/1.8$ or at 4.0°C , $R_T = 3.2\text{ k}\Omega$	C1
	hence $R/3.2 = 2.4/1.8$ $R = 4.3\text{ k}\Omega$	A1
7(b)	$R_T = 3.37\text{ k}\Omega$ or R_T is greater (than $3.2\text{ k}\Omega$)	B1
	$V^+ > V^-$	M1
	hence output is $+5.0\text{ V}$	A1

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Question	Answer	Marks
7(c)	correct LED symbol	B1
	two diodes shown connected, in parallel and with opposite polarities, between V_{OUT} and earth	M1
	diodes labelled to show correct polarities consistent with (b) (G pointing from V_{OUT} to earth and B pointing from earth to V_{OUT} if (b) correct)	A1

Question	Answer	Marks
8(a)	force per unit current	B1
	force per unit length (of wire)	B1
	current normal to (magnetic) field	B1
8(b)(i)	forces (on PQ and RS) are horizontal	B1
	(hence they create) no moment about the pivot	B1
	or	
	forces (on PQ and RS) are equal and opposite	(B1)
	(hence there is) no <u>net</u> force (on the two sections)	(B1)
8(b)(ii)	realisation of the need to apply moments	C1
	$BILx = mgy$	C1
	$B \times 2.7 \times 1.2 \times 10^{-2} \times 7.5 = 45 \times 10^{-6} \times 9.81 \times 8.8$	
	$B = 1.6 \times 10^{-2} \text{ T}$	A1

Question	Answer	Marks
9(a)	$0 \rightarrow t_1$ horizontal straight line at non-zero value of V_H and $t_3 \rightarrow t_4$ horizontal straight line at different non-zero V_H	B1
	$t_1 \rightarrow t_3$ straight diagonal line with negative gradient and graph line starts at $(0, V_0)$ and ends at $(t_4, -2V_0)$	B1
9(b)	$E = 0$ for $0 \rightarrow t_1$ and $t_3 \rightarrow t_4$	B1
	E is non-zero at all points between $t_1 \rightarrow t_3$	M1
	E has constant magnitude between $t_1 \rightarrow t_3$	A1

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Question	Answer	Marks
10(a)	$V_0 = \sqrt{2} \times V_{\text{r.m.s.}} = \sqrt{2} \times 9.9 \text{ (= 14 V)}$ and $\omega = 2\pi f = 2\pi \times 50 \text{ (= 314 rad s}^{-1}\text{)}$	C1
	$V = 14 \sin 314t$	A1
10(b)	enables (resonating) nuclei to be located	B1
	resonant frequency depends on magnetic field strength	B1
	Any one from: • non-uniform field is (accurately) calibrated • (non-uniform) field may be varied to enable detection in different positions • unique (magnetic) field strength/frequency at each point	B1
10(c)	$I = I_0 \exp(-\mu x)$	C1
	$I = I_0 [\exp(-\mu x)_{\text{bone}} \times \exp(-\mu x)_{\text{soft tissue}}]$	C1
	$I = I_0 [\exp(-2.9 \times 0.40) \times \exp(-0.92 \times 1.4)]$	
	$I/I_0 = 0.0865$	C1
	ratio/dB = $10 \lg 0.0865$ $= -11 \text{ dB}$	A1

Question	Answer	Marks
11(a)	discrete amount/quantum/packet of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(b)	mostly dark/dark background	B1
	coloured lines	B1
11(c)(i)	6	A1
11(c)(ii)	1. maximum photon energy = $13.6 - 0.85$ $(= 12.75 \text{ eV})$	C1
	maximum kinetic energy = $(13.6 - 0.85) - 5.6$ $= 7.2 \text{ eV}$	A1
	2. energy = hc/λ	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / [(13.6 - 0.85) \times 1.60 \times 10^{-19}]$	C1
	$= 9.8 \times 10^{-8} \text{ m}$	A1

Question	Answer	Marks
12(a)	fusion: two nuclei <u>combine</u> to form a (single) nucleus	B1
	fission: a (single) large nucleus <u>divides</u> to form (smaller) nuclei	B1
	Any one from: • fusion is initiated by (very) high temperatures • fission is initiated by neutron bombardment • resulting nuclei in fission are of similar size • (both processes) release energy • binding energy per nucleon increases • total binding energy increases • fission involves release of neutrons	B1
12(b)(i)	neutron	B1
12(b)(ii)	1. zero	A1
	2. $(4 \times 11.3290 \times 10^{-13}) - (2 \times 1.7813 \times 10^{-13}) - (3 \times 4.5285 \times 10^{-13})$	C1
	energy change = $45.316 \times 10^{-13} - 17.148 \times 10^{-13}$ = $2.82 \times 10^{-12} \text{ J}$	A1
12(b)(iii)	1.0 mol or N_A nuclei of each energy = $2.817 \times 10^{-12} \times 6.02 \times 10^{23}$ = $1.7 \times 10^{12} \text{ J}$	A1